

MACHINE LEARNING

1. Supervised learning

Linear Regression

Real world Example:

- ✓ Predicting House Prices

Problem Statement:

A real estate company wants to predict the price of houses in a city based on various features like:

- Size of the house (in square feet)
- Number of bedrooms
- Location rating
- Age of the house

The goal is to build a predictive model that can estimate the price of a house based on these features.

Solution:

- ✚ Collect data on past house sales.
- ✚ Preprocess the data (remove missing values, normalize features).
- ✚ Split data into training and testing sets (e.g., 80%-20%).
- ✚ Train the linear regression model on the training set.
- ✚ Predict prices on the test set.
- ✚ Evaluate performance using metrics like:
 - Mean Absolute Error (MAE)
 - Root Mean Square Error (RMSE)
 - R^2 Score (coefficient of determination)

Example Prediction:

If the trained model learns the following equation:

$$\text{Price} = 20 + 0.03(\text{Size}) + 5(\text{Bedrooms}) - 0.5(\text{Age})$$

Then for a house with:

- Size = 2000 sq.ft
- Bedrooms = 3
- Age = 5 years

$$\begin{aligned}\text{Predicted Price} &= 20 + 0.03(2000) + 5(3) - 0.5(5) \\ &= 20 + 60 + 15 - 2.5 = ₹92.5 \text{ Lakhs}\end{aligned}$$

Final Output:

The model predicts that the house price will be ₹92.5 lakhs based on its features.

2.Unsupervised Learning

K-means clustering

Real world Example:

- ✓ Customer Segmentation in a Shopping Mall

Problem Statement:

A shopping mall wants to understand its customers better to improve marketing strategies.

They have data about their customers, such as:

- ✚ Annual Income (₹)
- ✚ Spending Score (1–100) — how much they spend on average

The company wants to group customers into clusters (segments) with similar spending habits and income levels — without predefined labels.

This helps them:

- ✚ Identify high-value customers
- ✚ Target specific groups with personalized offers
- ✚ Improve overall sales and customer experience

Solution:

Let's say after applying K-Means (K=3), we get the following clusters:

Cluster	Description	Example Customers
Cluster 1	Low income, low spending	Customer 5
Cluster 2	Medium income, medium spending	Customer 2, 6
Cluster 3	High income, high spending	Customer 3, 4

3.Reinforcement Learning

Policy Gradient Methods

Real world Example:

- ✓ Autonomous Robot Navigation

Problem Statement:

- ❖ A company is developing a delivery robot that must navigate a warehouse and deliver packages to specific locations without colliding with obstacles or humans.
- ❖ The robot must learn an optimal policy — that is, a mapping from states (its position, nearby obstacles) to actions (move forward, turn left, turn right, stop) — to maximize its total reward (successful deliveries, fewer collisions, faster paths).
- ❖ There's no predefined training data, only experience from interactions with the environment.

Solution:

- Initialize the policy network with random weights θ .

- The network takes state inputs (robot's sensors, map data) and outputs probabilities of each possible action.
- Interact with the environment:
 - The robot moves based on the current policy.
 - It receives rewards for progress and penalties for collisions or delays.
- Collect trajectories (sequences of state, action, reward).
- Compute the total reward (Return) for each trajectory.
- Update policy parameters using the gradient ascent rule:
 - Increase the probability of actions that led to higher rewards.
- Repeat until the policy converges (robot successfully navigates the warehouse efficiently).

Result / Solution Outcome:

After many episodes of training:

- The robot learns to move efficiently to the delivery spot.
- It avoids obstacles automatically, because such actions received negative rewards.
- The learned policy network can generalize to new warehouse layouts.